

ADAPTIVE SOLUTIONS & ADVISORY GROUP

PROPOSED TECHNICAL SOLUTION ARCHITECTURE

QIAGEN Capacity Planning Platform

The proposed solution is based on a modern cloud-native data and analytics platform built on either Microsoft Fabric or Databricks, with Power BI serving as the enterprise reporting and dashboarding layer.

The architecture follows a layered approach that separates data ingestion, storage, transformation, semantic modeling, and reporting, ensuring scalability, maintainability, and future extensibility.

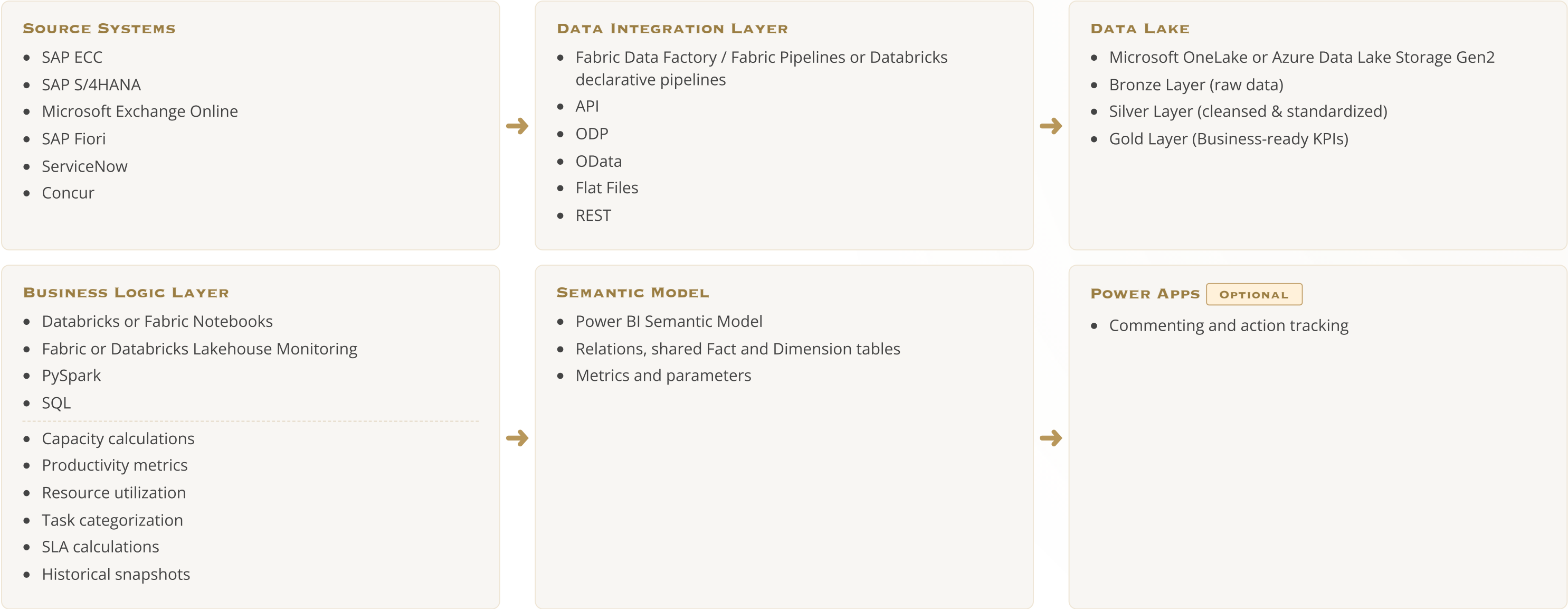
THE SOLUTION WILL

- ◆ Integrate data from SAP ECC, SAP S/4HANA, Microsoft Exchange Online, SAP Fiori, ServiceNow, and Concur.
- ◆ Support the coexistence of SAP ECC and SAP S/4HANA during the multi-year migration period.
- ◆ Centralize operational and productivity data into a single analytical platform.
- ◆ Calculate standardized KPIs and capacity planning metrics.
- ◆ Provide interactive dashboards for AP, AR, Team Leaders, Managers, and Finance Operations Leadership. Dashboards will have possibility to be exported to various formats and are read-only documents.
- ◆ Optionally **Not included in offer** allow users to enter comments, explanations, and action plans directly from dashboards using Microsoft Power Apps.
- ◆ Optionally **Not included in offer** to build DevOps integration with Dev, UAT and PROD environments.

The architecture is designed to support future onboarding of additional business units, countries, and operational systems with minimal changes.

High-level solution architecture

Logical architecture



END-TO-END DATA FLOW – EXTRACTION TO BUSINESS LOGIC

STEP 1 – DATA EXTRACTION

Operational data is extracted from multiple enterprise systems:

- ◆ SAP ECC
 - ◆ SAP S/4HANA
 - ◆ Exchange Online mailboxes
- ◆ SAP Fiori
 - ◆ ServiceNow
 - ◆ Concur

Extraction methods may include:

- ◆ Data Platform native connectors
 - ◆ SAP ODP
 - ◆ OData services
- ◆ REST APIs
 - ◆ others

Full snapshot loading from source systems.

Step 2 – Raw Data Landing (Bronze Layer)

Incoming datasets are stored unchanged in the Data Lake.

Step 3 – Data Standardization (Silver Layer)

Data is cleansed and standardized across all source systems. This layer creates a unified operational model independent of source system differences.

Step 4 – Business Logic (Gold Layer)

Business rules calculate standardized KPIs for capacity, workload, productivity and Resources management.

END-TO-END DATA FLOW – SEMANTIC MODEL & REPORTING

STEP 5 – SEMANTIC MODEL

A centralized Power BI semantic model exposes governed business entities.

Example dimensions, for example:

- ◆ Employee
- ◆ Team
- ◆ Company Code

Example fact tables:

- ◆ Tasks
- ◆ Capacity

The model enables consistent centralized KPI calculations across all dashboards.

Maximum of 10 facts and 15 dimensions in target model from source system directly transferred 1:1 from source systems. No complex aggregation algorithms to generate bronze and silver layer.

STEP 6 – REPORTING & SELF-SERVICE ANALYTICS

Preparing Power BI dashboards with underlying pages:

- ◆ **Capacity Dashboard** – As requested in RFP.
- ◆ **Management Dashboard – one-pager** **Optional** – Additional one page dashboard to support Managers decisions.
- ◆ **Operational KPI Dashboard - one-pager** **Optional** – Additional one page operational dashboard to track in details AP/AR performance.

Automated alerts may be set up on platform or semantic model level and can identify for example:

Capacity above 110%

Capacity below 70%

Backlog growth

SLA breaches

Significant workload changes

OPTIONAL POWER APPS INTEGRATION

OUT OF SCOPE

High-Level Solution Architecture

Technology Stack

Layer	Microsoft Fabric Option	Databricks Option
Data Integration	Fabric Data Factory, Fabric Pipelines	Azure Data Factory, Databricks Workflows
Data Storage	OneLake, Lakehouse	Azure Data Lake Storage Gen2, Delta Lake
Data Processing	Fabric Spark, Fabric Warehouse, Dataflows Gen2	Databricks Spark, Delta Live Tables, SQL Warehouse
Data Model	Fabric Warehouse / Lakehouse	Delta Tables + Star Schema
Business Logic	SQL, Spark, Dataflows	Spark, SQL, Python
Semantic Layer	Power BI Semantic Model	
Reporting	Power BI	
Dashboard Comments Optional	Power Apps + SharePoint	
Authentication	Microsoft Entra ID (Azure AD)	
Security	Role-Based Access Control (RBAC), Row-Level Security (RLS)	
Monitoring	Fabric Monitoring, Azure Monitor	Databricks Monitoring, Azure Monitor

BUSINESS DOMAINS

ACCOUNTS PAYABLE (AP)

SUPPORTED PROCESSES

The Accounts Payable scope may include:

- ◆ Invoice Processing
- ◆ Travel and Expense Processing
- ◆ Mailbox Management
- ◆ Intercompany Invoices and Recharges
- ◆ Month-End Closing
- ◆ Reporting
- ◆ Regular and Urgent Payments
- ◆ Vendor Master Data

KPIS AND METRICS

Candidate metrics may include:

Planned Capacity

Available Capacity

Capacity Utilization

Workload Volume

Processing Effort

Productivity Index

Average Processing Time

Backlog Volume

Backlog Aging and SLA Compliance

Relevant process-specific measures may also be defined for invoice processing, payments, mailbox activities, travel and expense processing, vendor master data, intercompany activities and month-end closing.

Detailed process boundaries, activity categories, workload drivers, calculation rules and KPI definitions – including formulas, thresholds and ownership – will be confirmed and validated with QIAGEN during Business Analysis.

CAPACITY PLANNING SCOPE

For the AP processes included in scope, the solution will support workload and capacity analysis by process, activity, task category, team and agreed organizational dimension – including incoming and processed workload, open workload, planned and available capacity, resource utilization, backlog and aging, absence and overtime adjustments, and workload and capacity trends.

SUPPORTED PROCESSES

The Accounts Receivable scope may include:

- ◆ Invoice Creation
- ◆ Cash Application
- ◆ Collections and Dispute Management
- ◆ Closing and Reconciliation
- ◆ SOX Controls and Reporting
- ◆ Adjustments including Write-Offs and Provisions
- ◆ Special Cases and Local Processes
- ◆ Credit Management

KPIS AND METRICS

Candidate metrics may include:

Planned Capacity

Available Capacity

Capacity Utilization

Workload Volume

Processing Effort

Productivity Index

Average Processing Time

Backlog Volume

Backlog Aging and SLA Compliance

Relevant process-specific measures may also be defined for invoices, cash application, unapplied cash, collections, disputes, credit management, closing, reconciliations, adjustments and SOX-related activities.

Detailed process boundaries, activity categories, calculation rules, granularity and KPI definitions – including formulas, thresholds and ownership – will be confirmed and validated with QIAGEN during Business Analysis.

CAPACITY PLANNING SCOPE

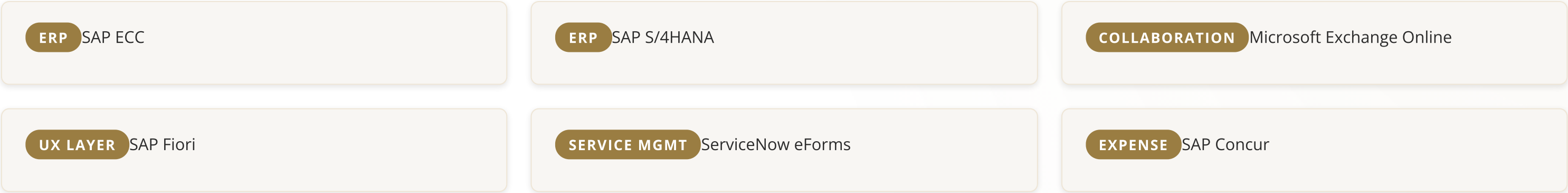
For the AR processes included in scope, the solution will support workload and capacity analysis by process, activity, task category, team and agreed organizational dimension – including incoming and processed workload, open workload, planned and available capacity, resource utilization, backlog and aging, absence and overtime adjustments, and workload and capacity trends.

The methodology will account for differences in workload drivers across invoice creation, cash application, collections, disputes, credit management and closing activities.

SOURCE SYSTEMS*

The proposed solution consolidates operational data from multiple enterprise applications into a single analytics platform. With maximum 10 facts and 10 dimensions.

The primary source systems include:



The solution is extensible, allowing additional systems or future SAP modules to be integrated as business requirements evolve.

COMMON DATA MODEL

To ensure consistent reporting across multiple operational systems, the solution introduces a centralized Common Data Model (CDM) that harmonizes and standardizes data from all source applications.

The Common Data Model eliminates differences in source system structures, naming conventions, and process implementations by transforming operational data into a unified business model.

Dimensions (example of)

◆ Employee

◆ Team

◆ Manager

◆ Site

◆ Company Code

◆ Country

◆ Process

◆ Task Category

◆ Task Type

◆ Organizational Unit

Fact Tables (example of)

◆ Tasks

◆ Workload

◆ Capacity

◆ Productivity

◆ Backlog

◆ Email Activity

◆ Absence

◆ Overtime

◆ SLA Performance

◆ Process Cycle Times

The Common Data Model also includes standardized business definitions for key metrics such as:

- Available Capacity
- Planned Capacity
- Capacity Utilization
- Processing Effort
- Productivity Index
- Task Aging
- Backlog Volume
- SLA Compliance
- Average Processing Time

By centralizing business definitions and KPI calculations, the Common Data Model ensures that all Power BI reports, dashboards, and analytical views are based on a single source of truth. This guarantees consistency across AP and AR teams, supports future scalability, and simplifies the addition of new source systems or business processes without requiring significant report redesign.

REPORTING AND ANALYTICS LAYER

The Reporting and Analytics Layer is implemented using Microsoft Power BI, providing a modern, interactive, and secure reporting environment for Finance Operations users across QIAGEN. Power BI consumes the centralized semantic model built on the curated Gold Layer, ensuring that all dashboards use consistent business definitions, standardized KPIs, and governed calculations.

The reporting solution includes role-based dashboards tailored to different user groups, including Finance Specialists, Team Leaders, Managers, and Executive Leadership. Core dashboards include:

RFP REQUESTED

Capacity Dashboard (with 3 pages)

- Overall workload distribution
- Capacity utilization
- Resource allocation
- Available vs. planned capacity
- Over- and under-utilization indicators

OPTIONAL · ONE-PAGER

Management Dashboard

Executive Summary

- Capacity
- Productivity
- Backlog
- SLA
- Trends

Workforce Planning

- Staffing simulation
- Resource utilization
- Capacity forecast
- Over/Under utilization

OPTIONAL · ONE-PAGER

Operational KPI Dashboard

AP Performance

- Invoice Processing
- Payments
- Vendor Master
- Mailbox

AR Performance

- Collections
- Cash Application
- Credit Management
- Closing

Power BI supports scheduled and automated data refreshes (daily, weekly, or monthly), self-service analysis with governed datasets, drill-through and cross-filtering, and export to PowerPoint or PDF. Combined with Row-Level Security (RLS), users only access data relevant to their organizational role, with a consistent and trusted view of Finance Operations performance across all regions.

SECURITY AND IDENTITY MANAGEMENT

The proposed solution leverages the Microsoft security ecosystem to provide secure authentication, authorization, and governance across all platform components. Access to data, reports, and planning applications is centrally managed using Microsoft Entra ID (formerly Azure Active Directory), integrating with QIAGEN's existing identity infrastructure and following the principle of least privilege so users only access what their role requires.

KEY SECURITY CAPABILITIES INCLUDE

Identity & Authentication

- ◆ Single Sign-On (SSO) using Microsoft Entra ID
- ◆ Multi-Factor Authentication (MFA) supported through corporate identity policies
- ◆ Integration with existing corporate user groups and security policies
- ◆ Service Principal authentication for automated data integration and scheduled processes

Authorization

- ◆ Role-Based Access Control (RBAC) across the data platform
- ◆ Row-Level Security (RLS) in Power BI to restrict data visibility based on organizational structure, business unit, site, or Company Code
- ◆ Workspace- and dataset-level permissions aligned with QIAGEN governance standards

Data Protection

- ◆ Encryption of data in transit (TLS) and at rest using platform-native encryption
- ◆ Secure storage of credentials and secrets using Azure Key Vault (or Microsoft Fabric Managed Secrets where applicable)
- ◆ Data masking and protection of sensitive employee information where required
- ◆ Compliance with QIAGEN security policies and applicable data privacy regulations (e.g., GDPR)

Governance & Auditing

- ◆ End-to-end data lineage and traceability
- ◆ Monitoring of data refreshes, pipeline executions, and security events
- ◆ Support for Microsoft Purview (optional) to enhance data governance, metadata management, and compliance reporting

This security architecture provides a secure and scalable foundation while minimizing administrative overhead and ensuring compliance with enterprise security standards.

INFRASTRUCTURE AND PLATFORM SERVICES

The solution is deployed on a modern, cloud-native analytics platform based on either Microsoft Fabric or Azure Databricks, leveraging Microsoft Azure as the underlying infrastructure – following a Platform-as-a-Service (PaaS) approach that lets QIAGEN focus on business value rather than infrastructure administration. Provisioning and maintenance of the cloud infrastructure remain on QIAGEN's side.

Key infrastructure and platform services include

Compute & Data Platform

- ◆ Fabric Lakehouse/Warehouse or Databricks with Delta Lake
- ◆ Elastic compute scaling with workload demand
- ◆ Separation of storage and compute for cost optimization

Data Storage

- ◆ OneLake (Fabric) or Azure Data Lake Storage Gen2
- ◆ Delta Lake format with ACID transactions and versioning
- ◆ Minimum 24-month historical retention, configurable archival

Orchestration & Automation

- ◆ Fabric Data Pipelines or Azure Data Factory
- ◆ Databricks Workflows for business logic (Databricks option)
- ◆ Automated refresh scheduling, retries and failure alerts

Monitoring & Operations

- ◆ Pipeline execution monitoring
- ◆ Performance and workload metrics
- ◆ Centralized logging for operational support

Scalability & Availability

- ◆ Cloud-native, scales with users and data volumes
- ◆ High availability via managed Azure services
- ◆ Elastic scaling for month-end/year-end peaks

DevOps & Deployment Optional

- ◆ CI/CD via Azure DevOps or Fabric Deployment Pipelines
- ◆ Dev / Test-UAT / Production environment separation
- ◆ Version-controlled, repeatable release management

Extensibility

More facts and dimensions

Incremental refresh mode

Additional ERP / Finance systems

Copilot / Databricks Genie

Predictive capacity planning

Anomaly detection and alerting

Expansion beyond AP and AR

This infrastructure provides a robust, scalable, and future-ready foundation supporting QIAGEN's global Finance Operations, ensuring performance, security and operational efficiency.

INTEGRATION DATA LAYERS – PRINCIPLES & BRONZE LAYER

The proposed Data Architecture follows the Medallion Architecture (Bronze, Silver, Gold) and is designed to provide a scalable, governed, and business-oriented data foundation for Capacity Planning and KPI reporting. The architecture supports multiple heterogeneous source systems while ensuring standardized business definitions, high data quality, and consistent reporting across all Finance Operations teams.

The architecture separates data ingestion, transformation, business logic, and reporting into distinct layers, enabling maintainability, auditability, and future extensibility.

THE KEY PRINCIPLES OF THE PROPOSED DATA ARCHITECTURE INCLUDE

- ◆ Single Source of Truth for all Capacity Planning and KPI reporting
- ◆ Standardized business definitions and KPI calculations
- ◆ Support for parallel SAP ECC and SAP S/4HANA environments
- ◆ Incremental data loading and historical data retention (minimum 24 months)
- ◆ Separation of raw, standardized, and business-ready datasets
- ◆ Scalability to onboard additional business processes and source systems
- ◆ Full data lineage and traceability across the entire data lifecycle

The solution adopts a layered data architecture that progressively transforms operational data into trusted analytical datasets.

BRONZE LAYER – RAW DATA Stores data extracted directly from operational source systems without business transformations.

Characteristics

- ◆ Raw source data preserved for auditability
- ◆ Historical snapshots maintained
- ◆ Incremental data ingestion
- ◆ Source-specific schemas retained
- ◆ Minimal transformations (technical metadata only)

Typical datasets examples

- ◆ SAP workflow records
- ◆ Invoice processing transactions
- ◆ Exchange mailbox activity
- ◆ Fiori absence records
- ◆ ServiceNow requests
- ◆ Concur transactions

INTEGRATION DATA LAYERS – SILVER & GOLD LAYER

SILVER LAYER – STANDARDIZED DATA Cleanses, validates, and standardizes data from all source systems into a unified operational model.

Typical transformations examples may include:

- ◆ Standardizing task categories
- ◆ Harmonizing employee identifiers
- ◆ Company Code mapping
- ◆ Date and time normalization
- ◆ Duplicate detection and removal
- ◆ Data type validation
- ◆ Business rule validation
- ◆ Organizational hierarchy enrichment
- ◆ Calendar enrichment

The Silver Layer removes source-system differences and establishes a consistent foundation for downstream reporting.

GOLD LAYER – BUSINESS DATA Contains business-ready datasets optimized for reporting, dashboards, and analytical workloads.

This layer may include

- ◆ Capacity Planning fact tables
- ◆ Productivity metrics
- ◆ Workload summaries
- ◆ Backlog analysis
- ◆ SLA performance
- ◆ Historical KPI snapshots
- ◆ Executive reporting datasets

Business calculations performed in this layer may include

- ◆ Capacity Utilization
- ◆ Available Capacity
- ◆ Effective Capacity
- ◆ Productivity Index
- ◆ Average Processing Time
- ◆ Task Aging
- ◆ Backlog Growth
- ◆ SLA Compliance
- ◆ Workload Allocation

High data quality is essential to ensure reliable capacity planning, operational reporting, and management decision-making. The proposed solution incorporates data quality controls throughout the entire data lifecycle, from ingestion to reporting.

Data quality validation is performed automatically during the data integration process, with issues logged and monitored for corrective action.

THE SOLUTION INCLUDES THE FOLLOWING DATA QUALITY CAPABILITIES

Data Validation

- ◆ Mandatory field validation
- ◆ Data type and format validation
- ◆ Business rule validation

Standardization

The solution standardizes data originating from different operational systems by:

- ◆ Harmonizing employee identifiers
- ◆ Mapping Company Codes and organizational structures
- ◆ Standardizing task categories and process names
- ◆ Normalizing date/time formats and time zones
- ◆ Applying consistent naming conventions across all datasets

Monitoring & Exception Handling

Monitoring identifies potential data quality issues, including:

- ◆ Duplicate transactions
- ◆ Invalid business values
- ◆ Delayed source system updates

Detected issues are logged, and automated notifications can be configured for operational support teams.

Data Lineage & Auditability

To support governance and troubleshooting, the solution maintains complete data lineage, including:

- ◆ Source system identification
- ◆ Extraction timestamps
- ◆ Transformation history
- ◆ Data refresh status
- ◆ Pipeline execution logs

BUSINESS ANALYSIS OBJECTIVE

Our Business Analysis approach is designed to translate QIAGEN's business needs, operational processes and management expectations into a structured, validated and implementation-ready set of requirements for the Capacity Planning solution.

Acting as the bridge between QIAGEN Finance Operations stakeholders and the technical implementation partner, we will ensure that the solution reflects the relevant ways of working across Accounts Payable and Accounts Receivable, applies consistent business definitions and provides reliable information for capacity and operational management decisions.

THE KEY OBJECTIVES OF THE BUSINESS ANALYSIS ARE TO

- ◆ establish a common understanding of the business objectives, expected outcomes and solution scope;
- ◆ understand the AP and AR processes relevant to capacity planning and reporting;
- ◆ identify workload drivers and current capacity planning practices;
- ◆ define the capacity planning and performance measurement framework required for the solution;
- ◆ identify and prioritize business, functional and reporting requirements;
- ◆ define consistent KPI definitions, calculation rules and business data requirements;
- ◆ identify relevant process, organizational and data variations;
- ◆ assess whether available business data can support the required use cases and calculations;
- ◆ provide clear and traceable business input to the technical implementation partner;
- ◆ support business validation throughout solution design, development and User Acceptance Testing.

The Business Analysis will focus on the requirements supporting the Capacity Planning Dashboard and any additional reporting components included in the final agreed solution scope.

ROLE OF THE BUSINESS ANALYSIS PARTNER

As the Business Analysis Partner, we will lead and coordinate the following activities in collaboration with QIAGEN business stakeholders and the technical implementation partner:

- ◆ business stakeholder engagement;
 - ◆ targeted business process and operational analysis;
 - ◆ assessment of current capacity planning and performance measurement practices;
 - ◆ identification and prioritization of business and functional requirements;
 - ◆ definition of workload, capacity and utilization concepts;
 - ◆ definition of KPI calculation logic and business rules;
 - ◆ preparation of business data requirements;
- ◆ documentation of relevant process and organizational variations;
 - ◆ definition of reporting requirements and acceptance criteria;
 - ◆ facilitation and documentation of business decisions;
 - ◆ business review of solution designs, mappings and dashboard prototypes;
 - ◆ support in preparing and coordinating the business aspects of User Acceptance Testing;
 - ◆ preparation of business documentation and knowledge transfer materials.

QIAGEN

QIAGEN's designated business representatives will remain responsible for approving business definitions, requirements, KPIs, solution designs and final business acceptance.

Technical Implementation Partner

The technical implementation partner will remain responsible for detailed source-system assessment, interface and extraction design, technical source-to-target mappings, implementation of transformation logic, semantic model development, dashboard development and technical testing.

We propose an iterative and evidence-based methodology combining business discovery, targeted process analysis, requirements engineering, data validation and continuous alignment between business and technical teams.

WORKSTREAM 1 – MOBILIZATION AND SCOPE ALIGNMENT

The objective of this workstream is to establish a shared understanding of the business case, expected outcomes, project boundaries and decision-making structure.

KEY ACTIVITIES

- ◆ Review the RFP, available documentation, existing reports and the proposed technical architecture.
- ◆ Confirm the AP and AR processes, organizational units, locations and user groups included in scope.
- ◆ Identify relevant Process Owners, Subject Matter Experts, report users, data owners and business approvers.
- ◆ Clarify the intended business use cases for operational teams, Team Leaders, Managers and Finance Operations Leadership.
- ◆ Confirm roles and responsibilities across QIAGEN, the Business Analysis Partner and the technical implementation partner.
- ◆ Identify key assumptions, dependencies, constraints and open business questions.
- ◆ Agree the requirements management, review and approval approach.

KEY OUTPUTS

- | | |
|--|---|
| ◆ confirmed business scope; | ◆ initial assumptions, dependencies and issues; |
| ◆ stakeholder and responsibility overview; | ◆ initial requirements catalogue. |
| ◆ Business Analysis plan; | |

WORKSTREAM 2 – CURRENT-STATE PROCESS AND CAPACITY PLANNING ASSESSMENT

The purpose of the current-state assessment is to understand how the relevant AP and AR work is received, categorized, allocated, processed, monitored and reported today. The assessment will be conducted to the level of detail required to define workload drivers, capacity rules, KPI logic, business data requirements and reporting functionality. It is not intended to constitute a comprehensive end-to-end process redesign or operating model review unless separately agreed.

KEY ACTIVITIES

- ◆ Relevant AP and AR processes and activities.
- ◆ Operational task and workload categories.
- ◆ Workload entry points and allocation practices.
- ◆ Roles and organizational structures relevant to capacity reporting.
- ◆ Current capacity planning and productivity measurement methods.
- ◆ Backlog, aging and SLA monitoring practices.
- ◆ Treatment of absence, overtime and selected non-operational activities.
- ◆ Existing operational and management reports.
- ◆ Manual calculations, adjustments and reconciliations.
- ◆ Relevant variations by team, country, Company Code, site or source system.
- ◆ Known reporting limitations and data-quality concerns.

Selected analysis techniques may include stakeholder interviews, facilitated workshops, document and report reviews, targeted process walkthroughs, process decomposition and fit-for-purpose process mapping.

KEY OUTPUTS

- ◆ current-state process and capacity planning summary;
- ◆ process and activity overview;
- ◆ workload-driver inventory;
- ◆ relevant process and reporting variations;
- ◆ existing KPI and report inventory;
- ◆ confirmed business scenarios to be supported by the solution.

WORKSTREAM 3 – BUSINESS REQUIREMENTS AND USE-CASE DEFINITION

This workstream will translate the agreed business objectives and operational needs into structured and prioritized solution requirements. Requirements will be defined from the perspective of the decisions and activities that the solution needs to support, rather than solely as a list of dashboard visualizations.

KEY ACTIVITIES

- ◆ Business and functional requirements.
- ◆ Reporting and visualization requirements.
- ◆ KPI and calculation requirements.
- ◆ Business data requirements.
- ◆ Filtering and drill-down requirements.
- ◆ Security and access requirements from a business perspective.
- ◆ Data-quality and reconciliation requirements.
- ◆ Historical reporting requirements.
- ◆ Acceptance criteria.

Requirements will be prioritized using an agreed method, such as Must Have, Should Have, Could Have and Won't Have for the current release. Prioritization will consider business value, data availability, technical feasibility, implementation complexity, dependencies and alignment with the agreed RFP scope.

KEY OUTPUTS

- ◆ prioritized Business and Functional Requirements Catalogue;
- ◆ business use cases;
- ◆ acceptance criteria;
- ◆ requirements traceability.

WORKSTREAM 4 – CAPACITY PLANNING AND KPI FRAMEWORK

A dedicated part of the analysis will define the capacity planning concepts, workload methodology, KPIs and business rules required for the solution.

Based on the agreed requirements and available data, the analysis may cover concepts such as:

- ◆ planned capacity;
- ◆ available and effective capacity;
- ◆ utilized and unallocated capacity;
- ◆ workload;
- ◆ productive and selected non-productive time;
- ◆ absence and overtime;
- ◆ processing effort;
- ◆ capacity utilization;
- ◆ productivity;
- ◆ backlog and aging;
- ◆ SLA performance.

For each agreed measure, we will define, as applicable:

- ◆ business definition and purpose;
- ◆ formula and unit of measure;
- ◆ numerator and denominator;
- ◆ calculation period and granularity;
- ◆ inclusion and exclusion rules;
- ◆ treatment of missing, incomplete or corrected data;
- ◆ treatment of absence, overtime and shared resources;
- ◆ allocation and aggregation rules;
- ◆ applicable thresholds;
- ◆ required data and expected source;
- ◆ responsible business owner;
- ◆ acceptance criteria.

The workload methodology will distinguish transaction volume from processing effort where this distinction is relevant and supported by available data. Any requirement for advanced complexity weighting, predictive forecasting, optimization algorithms or custom planning engines will be jointly assessed with the technical implementation partner and is **not included in the baseline scope** unless explicitly agreed.

The resulting KPI Catalogue will serve as the business reference for the implementation of calculation logic within the data model, semantic layer and dashboards.

KEY OUTPUTS

- ◆ Capacity Planning Framework;
- ◆ business-approved KPI Catalogue;
- ◆ exception and threshold rules;
- ◆ KPI ownership and KPI-to-source traceability.

WORKSTREAM 5 – BUSINESS DATA REQUIREMENTS AND BUSINESS-TO-TECHNICAL ALIGNMENT

The Business Analysis team will define the information required to support the approved processes, use cases, KPIs and dashboards. This activity will provide the business context for technical source-system assessment, data mapping and transformation design. It will not replace the detailed technical analysis performed by the technical implementation partner.

KEY ACTIVITIES

- ◆ Identify the business entities and attributes required by the solution.
- ◆ Define business terminology and relationships between key entities.
- ◆ Specify the data required for calculations, filtering, reporting and reconciliation.
- ◆ Define expected data granularity and historical coverage.
- ◆ Identify relevant organizational and reporting hierarchies.
- ◆ Document known data gaps and quality concerns.
- ◆ Define business validation and reconciliation rules.
- ◆ Align business requirements with the proposed Common Data Model.
- ◆ Review technical mappings and transformation rules from a business perspective.
- ◆ Validate that implemented calculations and transformations remain consistent with approved business definitions.

Each requirement and KPI will be assessed jointly with the technical implementation partner against source-data availability, data granularity, historical coverage, data quality and technical feasibility. Where the available data does not support a requested business outcome, the limitation and possible alternatives will be documented for business decision.

The analysis will also consider relevant differences associated with the coexistence of SAP ECC and SAP S/4HANA and will support the definition of system-independent business rules where feasible.

KEY OUTPUTS

- ◆ Business Data Requirements Specification;
- ◆ business entity and attribute definitions;
- ◆ data-quality and reconciliation requirements;
- ◆ traceability between business requirements, KPIs, data and reports.

WORKSTREAM 6 – REPORTING AND USER EXPERIENCE DESIGN

Reporting requirements will be driven by user decisions and business outcomes while taking into account data availability, data quality, technical feasibility and the agreed reporting scope.

KEY ACTIVITIES

- ◆ Identify user groups and their information needs.
- ◆ Define the purpose and intended business decisions supported by each dashboard page.
- ◆ Establish the required operational and management views.
- ◆ Define dashboard hierarchy and navigation.
- ◆ Specify filters, slicers, drill-down and drill-through requirements.
- ◆ Define default views and reporting periods.
- ◆ Determine the appropriate level of detail for each user group.
- ◆ Prepare low-fidelity wireframes or reporting mock-ups.
- ◆ Facilitate iterative prototype reviews.
- ◆ Validate the developed reports against approved requirements and KPI definitions.

Reporting requirements, wireframes and prototypes will be developed within the agreed baseline, including the confirmed number of dashboard pages, visualizations, user roles and supported Power BI interactions.

KEY OUTPUTS

- ◆ Dashboard Functional Design;
- ◆ dashboard content and navigation requirements;
- ◆ wireframes or mock-ups;
- ◆ report-to-KPI mapping;
- ◆ business terminology and interpretation guidance.

WORKSTREAM 7 – BUSINESS VALIDATION AND USER ACCEPTANCE TESTING

Business validation will be embedded throughout the delivery lifecycle. Key requirements, KPI definitions, data rules, wireframes and prototypes will be reviewed with designated stakeholders before final implementation.

We will support QIAGEN in defining, preparing and coordinating the business aspects of User Acceptance Testing.

KEY ACTIVITIES

- ◆ Define business UAT scope and acceptance criteria.
- ◆ Prepare representative business scenarios and test cases.
- ◆ Define expected business results.
- ◆ Validate calculations against approved KPI rules.
- ◆ Reconcile selected dashboard results with agreed source information or reference reports.
- ◆ Validate filters, drill-downs and reporting views.
- ◆ Support business assessment and prioritization of identified issues.
- ◆ Support business retesting following technical corrections.
- ◆ Prepare the business acceptance recommendation and supporting documentation.

QIAGEN business representatives will execute or formally validate the agreed UAT scenarios and provide final business acceptance. The technical implementation partner will remain responsible for technical testing and the resolution of technical defects.

KEY OUTPUTS

- ◆ business UAT scenarios and expected results;
- ◆ KPI reconciliation criteria and evidence;
- ◆ requirements coverage summary;
- ◆ business acceptance recommendation.

Business Analysis governance will be integrated into the overall project governance model and aligned with the technical delivery cadence.

Stakeholder engagement will include:

- ◆ Business Analysis working sessions with Process Owners, Subject Matter Experts, reporting users and data representatives;
 - ◆ joint business and technical alignment sessions;
 - ◆ design and prototype reviews;
- ◆ documented decision, issue and dependency management;
 - ◆ escalation of unresolved business matters through the agreed project governance structure.

Key business definitions, requirements and solution components will be reviewed and approved by designated QIAGEN representatives in accordance with the agreed governance and approval process.

KEY BUSINESS ANALYSIS DELIVERABLES

The proposed Business Analysis deliverables will be consolidated into the following packages:

- 1

Business Analysis and Discovery Summary – Confirmed business scope, stakeholder landscape, current-state findings, process and activity overview, key variations, assumptions and constraints.
- 2

Business and Functional Requirements Specification – Prioritized requirements, use cases, reporting and access requirements, acceptance criteria and requirements traceability.
- 3

Capacity Planning and KPI Catalogue – Capacity and workload definitions, KPI formulas, business rules, calculation granularity, inclusions, exclusions, exceptions, thresholds and ownership.
- 4

Business Data Requirements Specification – Required business entities and attributes, business definitions, granularity, historical requirements, data-quality controls, reconciliation rules and links to KPIs and reports.
- 5

Dashboard Functional Design – User needs, dashboard content, filters, navigation, drill-down requirements, wireframes and report-to-KPI mapping.
- 6

Business Validation and UAT Package – Business test scenarios, expected results, KPI reconciliation criteria, requirements coverage and the business acceptance recommendation.

Supporting artefacts, including the RACI, business glossary, decision log, issue log and traceability matrix, will be maintained as part of the relevant

SOURCE SYSTEM ASSESSMENT

The objective of the Source System Assessment is to evaluate all systems that will provide data for the Capacity Planning solution and determine the optimal integration strategy for each of them.

THE ASSESSMENT WILL COVER

System Inventory

Identification of all relevant source systems, including:

- ◆ SAP ECC
 - ◆ SAP S/4HANA
 - ◆ Microsoft Exchange Online
- ◆ SAP Fiori
 - ◆ ServiceNow
 - ◆ SAP Concur

Technical Assessment

For each source system, the following aspects will be evaluated:

- ◆ Available integration methods (APIs, OData, JDBC, reports, flat files)
- ◆ Data ownership
- ◆ Available business objects
- ◆ Data refresh capabilities
- ◆ Historical data availability
- ◆ Performance considerations
- ◆ Authentication mechanisms
- ◆ Existing reporting interfaces
- ◆ Planned system changes (e.g., SAP ECC to SAP S/4HANA migration)

Business Assessment

The implementation team will identify:

- ◆ Business processes supported by each system
- ◆ Process owners
- ◆ Available operational metrics
- ◆ Existing reporting limitations
- ◆ Required KPIs
- ◆ Data completeness and quality

The outcome of this activity will be a Source System Assessment Report documenting integration recommendations and implementation priorities.

INTERFACE ANALYSIS

The Interface Analysis focuses on identifying and documenting all required interfaces between the source systems and the Capacity Planning platform.

For each interface, the following information will be defined:

Interface Characteristics

- ◆ Source system
- ◆ Target platform
- ◆ Integration technology
- ◆ Data format
- ◆ Communication protocol
- ◆ Authentication method
- ◆ Refresh frequency
- ◆ Expected data volumes
- ◆ Incremental vs. full extraction

Technical Specifications

Each interface specification will include:

- ◆ Source objects (tables, APIs, reports)
- ◆ Required fields
- ◆ Filtering criteria
- ◆ Extraction logic
- ◆ Scheduling requirements
- ◆ Error handling procedures
- ◆ Retry mechanisms
- ◆ Monitoring requirements

Special attention will be given to the coexistence of SAP ECC and SAP S/4HANA, ensuring that interface designs remain stable throughout the ERP migration and minimize future rework.

DATA MAPPING

The Data Mapping activity defines how operational data from multiple source systems is transformed into the Common Data Model used for reporting and analytics.

The objective is to establish consistent business definitions across all Finance Operations processes.

THE MAPPING EXERCISE INCLUDES

Business Entity Mapping

Definition of mappings for core business entities such as:

- ◆ Employees
- ◆ Teams
- ◆ Company Codes
- ◆ Sites
- ◆ Organizational Units
- ◆ Task Categories
- ◆ Finance Processes
- ◆ Calendar

Operational Data Mapping

Mapping of operational data elements including:

- ◆ Task identifiers
- ◆ Workflow statuses
- ◆ Processing timestamps
- ◆ Processing effort
- ◆ Email activity
- ◆ Leave and absence records
- ◆ Overtime
- ◆ Service requests
- ◆ Travel expense processing

KPI Mapping

Business rules for KPI calculations will be documented, including:

- ◆ Capacity Utilization
- ◆ Available Capacity
- ◆ Planned Capacity
- ◆ Productivity Index
- ◆ Average Processing Time
- ◆ Backlog Volume
- ◆ SLA Compliance
- ◆ Task Aging
- ◆ Workload Distribution

TECHNICAL DISCOVERY

The Technical Discovery phase validates the technical feasibility of the proposed solution and identifies any implementation constraints before development begins.

THE ACTIVITIES INCLUDE

Platform Readiness Assessment

Verification of the selected technology platform, including:

- ◆ Microsoft Fabric or Azure Databricks environment readiness
- ◆ Power BI tenant configuration
- ◆ Identity and access management
- ◆ Workspace and environment setup
- ◆ Connectivity to source systems

Connectivity Validation

Technical validation of:

- ◆ SAP connectivity
- ◆ Microsoft Graph API access
- ◆ ServiceNow API access
- ◆ Concur API access
- ◆ Network connectivity
- ◆ Firewall requirements
- ◆ Authentication mechanisms

Data Volume Assessment

Analysis of:

- ◆ Expected transaction volumes
- ◆ Historical data availability
- ◆ Growth projections
- ◆ Data refresh windows
- ◆ Storage requirements
- ◆ Performance expectations

Technical Constraints Assessment

Identification of potential implementation constraints, including:

- ◆ Source system limitations
- ◆ API throttling limits
- ◆ Extraction performance
- ◆ Data availability windows
- ◆ Infrastructure dependencies
- ◆ Security and compliance requirements

The Functional Design defines how the solution will support Finance Operations processes and satisfy the business requirements described in the RFP.

The design is developed in close collaboration with Finance Operations stakeholders through workshops and iterative review sessions.

THE FUNCTIONAL DESIGN COVERS

Business Processes

Definition of the supported business processes, including:

- ◆ Accounts Payable (AP)
- ◆ Accounts Receivable (AR)
- ◆ Invoice Processing
- ◆ Payments
- ◆ Vendor Master Data
- ◆ Mailbox Management
- ◆ Travel & Expense Processing
- ◆ Month-End Closing (MEC)
- ◆ Reporting activities
- ◆ Other operational support processes

Functional Requirements

Specification of:

- ◆ Capacity Planning functionality
- ◆ Resource utilization analysis
- ◆ Productivity measurement
- ◆ Workload allocation
- ◆ Historical trend analysis
- ◆ Backlog monitoring
- ◆ SLA reporting
- ◆ Alert generation
- ◆ Dashboard navigation and drill-down capabilities

KPI Catalogue

Definition of all business KPIs, including:

- ◆ Capacity Utilization
- ◆ Available Capacity
- ◆ Planned Capacity

Reporting Design

Definition of the reporting structure may include:

- ◆ Capacity Planning Dashboard – 3 pages
- ◆ Management Dashboard (optional) – 1 page
- ◆ Operational KPI Dashboard (optional) – 1 page

TECHNICAL DESIGN

The Technical Design defines how the solution will be implemented using Microsoft Fabric or Azure Databricks together with Power BI.

The design follows modern cloud architecture principles based on scalability, modularity, and maintainability.

THE TECHNICAL DESIGN INCLUDES

Platform Architecture

- ◆ Microsoft Fabric or Azure Databricks architecture
- ◆ Lakehouse / Delta Lake architecture
- ◆ Compute and storage services
- ◆ Environment strategy (extend to Development, Test, Production as an additional option)
- ◆ Capacity sizing considerations

Integration Design

- ◆ Source system connectors
- ◆ Data ingestion mechanisms
- ◆ API integrations
- ◆ Loading strategy
- ◆ Scheduling
- ◆ Pipeline orchestration
- ◆ Error handling

Data Processing Design

- ◆ Bronze, Silver and Gold data layers
- ◆ Transformation workflows
- ◆ Business rules implementation
- ◆ Historical snapshot strategy
- ◆ KPI calculation processes

Reporting Architecture

- ◆ Power BI Semantic Model
- ◆ Shared dataset
- ◆ DAX calculation layer
- ◆ Row-Level Security implementation if requested
- ◆ Dashboard architecture
- ◆ Power Apps integration

Out of scope

Security Design ensures that the solution complies with QIAGEN's security policies while protecting operational and employee-related information.

The design follows Microsoft security best practices and applies the principle of least privilege.

THE SECURITY DESIGN INCLUDES

Identity Management

- ◆ Microsoft Entra ID integration
- ◆ Single Sign-On (SSO)
- ◆ Multi-Factor Authentication (MFA)
- ◆ Service Principal authentication for automated processes

Access Management

Definition of:

- ◆ Role-Based Access Control (RBAC)
- ◆ Power BI Row-Level Security (RLS) if requested
- ◆ Workspace permissions
- ◆ Dataset permissions
- ◆ Administrative roles

Access can be restricted based on:

- ◆ User role
- ◆ Department
- ◆ Team
- ◆ Company Code
- ◆ Site
- ◆ Management hierarchy

Data Protection

Specification of:

- ◆ Encryption in transit (TLS)

Auditing & Compliance

Definition of:

- ◆ User activity logging

DATA TRANSFORMATION

Following data ingestion, operational data is transformed into a standardized analytical model suitable for Capacity Planning and KPI reporting.

Transformations are implemented within Silver and Gold layers of the Medallion Architecture.

TYPICAL TRANSFORMATION ACTIVITIES INCLUDE

Data Standardization

- ◆ Standardization of task categories
- ◆ Harmonization of employee identifiers
- ◆ Company Code mapping
- ◆ Organizational hierarchy mapping
- ◆ Process classification
- ◆ Date and time normalization
- ◆ Time zone conversion where applicable

Business Rule Implementation

Implementation of business logic for:

- ◆ Capacity calculations
- ◆ Productivity metrics
- ◆ Workload allocation
- ◆ Task effort calculations
- ◆ Backlog determination
- ◆ SLA calculations
- ◆ Resource utilization
- ◆ Historical KPI snapshots

Common Data Model Population

Operational data from all source systems is transformed into standardized business entities, may include:

- ◆ Employees
- ◆ Teams
- ◆ Tasks
- ◆ Processes
- ◆ Capacity
- ◆ Productivity
- ◆ Absence
- ◆ Email activity

This transformation layer removes source-system-specific complexities and provides a unified foundation for reporting and analytics.

DATA VALIDATION

Data Validation ensures that the integrated data is complete, accurate, and suitable for business reporting before it is made available to end users.

Validation controls are embedded within the data pipelines and executed automatically during each data refresh.

THE VALIDATION PROCESS INCLUDES

Technical Validation

Verification of:

- ◆ Successful data extraction
- ◆ Record counts
- ◆ Mandatory fields
- ◆ Data types and formats
- ◆ Load completeness
- ◆ Duplicate records
- ◆ Referential integrity

Business Validation

Validation of business rules, including:

- ◆ Correct mapping of task categories
- ◆ Valid organizational assignments
- ◆ Capacity calculation consistency
- ◆ KPI calculation accuracy
- ◆ Processing time logic
- ◆ Workload allocation rules

Processing historical data that came from other source systems than those defined in RFP is out of scope.

REPORTING DEVELOPMENT

POWER BI DEVELOPMENT

The Power BI reporting solution will provide a centralized analytics environment for Finance Operations, enabling users to monitor operational performance, identify capacity constraints, and make data-driven resource planning decisions.

The solution will leverage a shared semantic model to ensure consistent KPI definitions and calculations across all reports.

SEMANTIC MODEL DEVELOPMENT

A centralized Power BI Semantic Model will be developed using a dimensional Star Schema.

Example of Dimensions

◆ Employee

◆ Team

◆ Company Code

◆ Site

◆ Process

◆ Task Type

◆ Calendar

◆ Source System

Example of Fact Tables

◆ Task Activity

◆ Capacity

◆ Productivity

◆ Email Activity

◆ Absence

◆ Overtime

◆ Backlog

◆ SLA Performance

The semantic model will expose reusable business measures, enabling consistent reporting across all dashboards.

KPI DEVELOPMENT

Standardized business measures will be implemented using DAX and centralized business logic. Examples include:

Capacity Utilization (%)

Available Capacity

Planned Capacity

Resource Utilization

Productivity Index

Average Processing Time

Tasks Completed

Open Tasks

Backlog Volume

Backlog Aging

SLA Compliance

Response Time

Overtime Hours

Workload Distribution by Task Type

Where appropriate, calculations will support historical trend analysis, period-over-period comparisons, and drill-down capabilities.

USER EXPERIENCE AND INTERACTIVITY

Reports will be designed following Power BI and Microsoft Fluent Design best practices to provide a consistent and intuitive user experience. Features include:

- ◆ Interactive filtering and slicing

◆ Drill-down and drill-through navigation

◆ Cross-report navigation

◆ Dynamic tooltips
- ◆ Bookmarks for predefined analytical views

◆ Responsive layouts

◆ Export to PowerPoint, PDF, and Excel (subject to Power BI capabilities and permissions)

SECURITY AND GOVERNANCE

The reporting layer will implement enterprise-grade security mechanisms, including:

- ◆ Row-Level Security (RLS) if requested

◆ Workspace-based access management

◆ Centralized semantic model governance
- ◆ Microsoft Entra ID authentication

◆ Certified and governed datasets

◆ Audit logging and usage monitoring

TESTING AND USER ACCEPTANCE

Reporting development will follow an iterative delivery approach with continuous business validation. Activities include:

- ◆ Testing of KPIs and DAX calculations

◆ Dashboard functional testing

◆ Performance testing
- ◆ User Acceptance Testing (UAT)

◆ Feedback incorporation and report refinement

◆ Final business sign-off prior to production deployment

This approach ensures that the reporting solution is accurate, user-friendly, and aligned with QIAGEN's operational objectives while providing a scalable foundation for future reporting enhancements.

Go-Live and Hypercare

Following the successful completion of all User Acceptance Testing (UAT) activities and formal business sign-off, the end-to-end Capacity Planning solution will be prepared for production deployment. A detailed Go-Live checklist will be executed to ensure a controlled transition from the test environment to the production environment with minimal disruption to business operations.

Deployment Activities

User communication materials will be prepared in advance, and the Power BI reports will be published and shared with the target user groups according to the defined security model. A Frequently Asked Questions (FAQ) document and dedicated support sessions will also be provided to address the most common user questions during the rollout period.

User Training - Knowledge Transfer

Comprehensive knowledge transfer sessions will be delivered for both the IT support team responsible for maintaining the solution and the business users who will consume the reports. Training will cover solution functionality, navigation, report interpretation, basic troubleshooting, and operational procedures to ensure successful adoption and long-term sustainability of the solution.

OUT OF SCOPE

Post-Go-Live Support

Post-Go-Live application support, including both technical and business support services, can be provided as an optional managed service following the successful implementation of the solution. The scope, service levels (SLAs), duration, and commercial terms of such support will be agreed separately as part of a dedicated support and maintenance agreement.

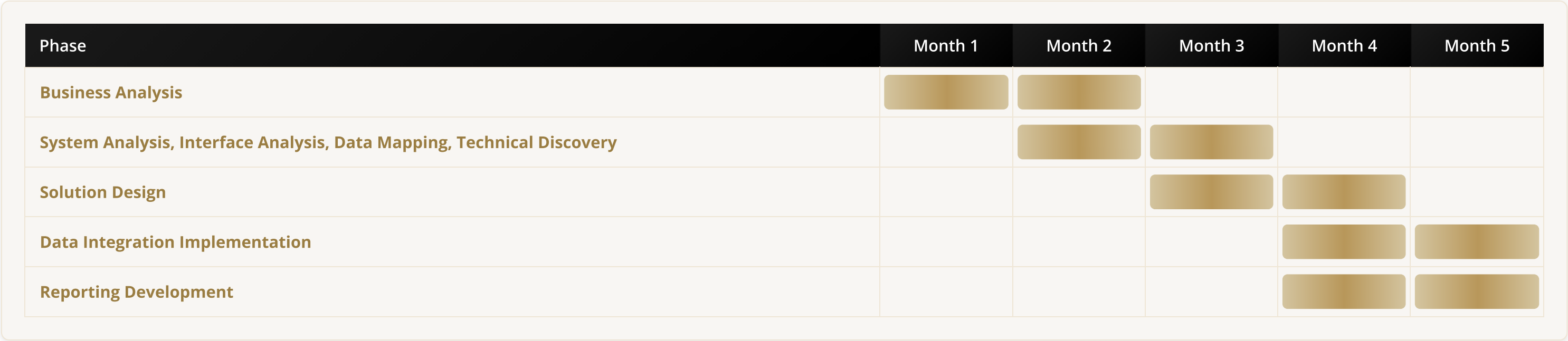
BUSINESS ASSUMPTIONS

The proposed Business Analysis scope and effort are based on the following assumptions:

- ◆ QIAGEN will confirm the AP and AR processes, organizational units, countries, Company Codes, teams and reporting users included in scope during project mobilization.
- ◆ PII (Personal Identification Information) to be managed according to company guidelines and regulatory regulations.
- ◆ The Business Analysis will be performed to the level of detail required to define workload drivers, capacity rules, KPI logic, business data requirements and dashboard functionality. Comprehensive process redesign, operating model redesign and organizational restructuring are outside the baseline scope unless separately agreed.
- ◆ QIAGEN will nominate appropriate Business Owners, Process Owners, Subject Matter Experts, Data Owners, report users and business approvers and ensure their availability for workshops, reviews, decision-making and User Acceptance Testing.
- ◆ QIAGEN will provide available process documentation, activity inventories, existing reports, KPI definitions, organizational information and representative business data required for the analysis.
- ◆ QIAGEN will remain responsible for approving the business scope, requirements, KPI definitions, business rules, dashboard designs and final business acceptance.
- ◆ Business requirements and KPIs will be assessed jointly against business value, source-data availability, data quality, historical coverage, technical feasibility and the agreed implementation scope.
- ◆ Where teams or locations apply different process or KPI definitions, QIAGEN will approve the common definition, required variation or exclusion to be implemented.
- ◆ Capacity, workload, utilization, productivity, backlog and SLA measures will be based on business-approved definitions and data available within the agreed source systems.
- ◆ The use of standard effort, task-complexity factors or alternative workload drivers will depend on the availability, reliability and business acceptance of the underlying data
- ◆ Advanced workload optimization, predictive forecasting, machine learning, simulation engines and complex custom planning algorithms are outside the baseline scope unless explicitly agreed.
- ◆ Business Analysis will identify data gaps and quality concerns relevant to the agreed use cases but will not include comprehensive data cleansing, manual data correction, master data remediation or reconstruction of unavailable historical data.
- ◆ QIAGEN will provide or approve the organizational hierarchies, employee-to-team assignments and business access rules required for reporting. Employee-level reporting will be subject to QIAGEN's privacy, security and access-governance requirements.
- ◆ Dashboard requirements and wireframes will be developed within the agreed reporting baseline, including the confirmed number of dashboards, pages, visualizations, user groups and supported Power BI interactions.
- ◆ The technical implementation partner will remain responsible for detailed source-system analysis, interface and extraction design, technical mappings, transformation implementation, semantic model development, dashboard development and technical testing.
- ◆ QIAGEN business representatives will execute or formally validate the agreed UAT scenarios and provide final business acceptance. The Business Analysis Partner will support business test preparation, KPI reconciliation, issue assessment and business retesting.
- ◆ Requirements or material changes introduced after approval of the relevant baseline may be subject to impact assessment and formal change control.
- ◆ Delays in stakeholder availability, required documentation, data access, business decisions or approvals may affect the agreed delivery schedule and will be managed through the project governance process.

ASSUMPTIONS

PROJECT GANTT CHART



Phase and month-column labels reproduced verbatim from the source Gantt chart (Month 1 – Month 5). Exact bar durations were not encoded as text in the source document; bars above are an illustrative rendering of the five-phase sequence.

INTEGRATION ASSUMPTIONS

The proposed effort is based on the following assumptions regarding data integration and transformation. Access to all required source systems and data sources will be provided by QIAGEN. QIAGEN is responsible for ensuring that the project team receives the necessary permissions and access rights to extract and access the required data. In the event that access is unavailable or insufficient due to internal authorization or privilege issues, QIAGEN will make reasonable efforts to resolve the issue within one (1) business day. Any delays resulting from unresolved access or permission issues may impact the agreed project timeline. If such delays extend beyond the agreed resolution period, the project schedule and any associated additional effort or costs will be reviewed and agreed upon by both parties through the project governance process.

Integration Logic

- ◆ Data integration will primarily use one-to-one mappings based on existing business or technical identifiers available in the source systems.
- ◆ Record matching between source systems will rely on existing keys using straightforward join logic (e.g., Full Outer Join where applicable).
- ◆ No fuzzy matching, probabilistic matching, or AI-based record linkage is included.

Data Processing

- ◆ Silver Layer transformations will be based on detailed transactional data without advanced aggregation or complex calculation logic.
- ◆ Gold Layer datasets will be generated primarily through straightforward business transformations and KPI calculations based on the curated Silver Layer.
- ◆ No advanced analytical algorithms, forecasting models, optimization engines, or machine learning models are included.

Historical Data Processing

- ◆ Historical data will be loaded using the same transformation logic as current data.
- ◆ A Full Snapshot loading approach is assumed.
- ◆ Separate historical processing algorithms (e.g., different business logic before and after a specific cut-off date) are out of scope.

Data Model Scope

- ◆ The implementation assumes an agreed and fixed number of business objects – max 10 fact tables and 10 datasets, one final semantic model, one main dashboard requested with RFP and optionally 2 additional one pager dashboards proposed with the offer.
- ◆ Any significant increase in the number of integrated entities or business objects may require additional implementation effort.

ADVANCED BUSINESS LOGIC, DATA CLEANSING, SECURITY & REPORTING ASSUMPTIONS

Advanced Business Logic

- ◆ The proposed implementation assumes standard KPI calculations and business rules.
- ◆ Complex business-specific calculation engines, advanced aggregation logic, optimization algorithms, or custom planning algorithms are not included unless explicitly defined during the requirements phase.
- ◆ If additional complex calculation algorithms are requested, they will be assessed separately and will require additional implementation effort.

Data Cleansing

- ◆ Only the data quality rules explicitly described in the Data Quality section are included.
- ◆ Data cleansing activities are limited to record-level validation, standardization, and enrichment based on existing source system attributes.
- ◆ Advanced data cleansing activities, manual data corrections, master data harmonization projects, or complex cross-system reconciliation processes are out of scope.

Security Assumptions

- ◆ Infrastructure security, network security, firewalls, VPN connectivity, and platform administration remain the responsibility of QIAGEN.
- ◆ Identity management and authentication will be provided through QIAGEN's existing Microsoft Entra ID (Azure Active Directory) environment.
- ◆ The proposed solution implements standard Role-Based Access Control (RBAC) and Power BI Row-Level Security (RLS) as described in the Security Design section.
- ◆ Advanced security features such as data anonymization, pseudonymization, tokenization, data masking beyond standard platform capabilities, or encryption key management customization are out of scope, unless explicitly requested.
- ◆ Compliance with QIAGEN's internal security policies and access governance processes is assumed.

Reporting Assumptions

- ◆ Development of one centralized Power BI semantic model supporting all reports.
- ◆ Development of one requested in RFP Power BI dashboard for read-only/export only interactions.
- ◆ Optionally development of 2 additional one pager dashboards.
- ◆ Maximum 25 report visualizations (charts, KPI cards, matrices, tables and maps where applicable).
- ◆ Responsive report design for desktop usage; dedicated mobile layouts are out of scope.
- ◆ Advanced self-service reporting capabilities, ad hoc report builders, paginated reports, or pixel-perfect reporting are out of scope.
- ◆ Any additional reports, dashboards, KPIs, or significant design changes requested after the Functional Design sign-off may require a formal change request and additional implementation effort.

INTEGRATION RISKS

The solution requires integration with multiple heterogeneous source systems, including SAP ECC, SAP S/4HANA, Microsoft Exchange Online, SAP Fiori, ServiceNow, and Concur.

Potential integration risks include

- ◆ Differences in data structures between SAP ECC and SAP S/4HANA during the migration period
- ◆ Limited availability of APIs or extraction mechanisms for certain applications
- ◆ Changes in source system interfaces during SAP transformation activities
- ◆ Performance impact caused by extracting large operational datasets
- ◆ Dependency on external system owners for access configuration and approvals

MITIGATION STRATEGY

Flexible Integration Architecture

- ◆ Use standardized enterprise integration patterns based on APIs, OData, SAP connectors, REST interfaces, and scheduled extraction mechanisms
- ◆ Separate source-specific ingestion logic from business transformation logic
- ◆ Implement a modular ingestion framework allowing individual sources to evolve independently

SAP ECC / SAP S/4HANA Coexistence Support

- ◆ Design a common data model independent of the source ERP system
- ◆ Maintain unified business definitions across ECC and S/4HANA
- ◆ Enable parallel ingestion from both environments during the migration period
- ◆ Avoid dependency on a single SAP version

Controlled Data Extraction

- ◆ Implement incremental loading where supported
- ◆ Schedule extraction activities outside critical operational periods
- ◆ Monitor extraction performance and optimize data volumes

Testing and Validation

- ◆ Perform integration testing with each source system
- ◆ Validate extracted data against operational reports
- ◆ Establish reconciliation processes with business stakeholders

DATA QUALITY RISKS

Capacity planning and productivity dashboards rely on accurate and complete operational data.

Potential data quality risks include

- ◆ Missing or incomplete task information
- ◆ Inconsistent task categorization across teams
- ◆ Different definitions of processing times or productivity metrics
- ◆ Incorrect employee mappings
- ◆ Inconsistent organizational structures between systems

MITIGATION STRATEGY

Common Data Model

- ◆ Implement a centralized business model with standardized definitions
- ◆ Harmonize task categories across AP and AR processes
- ◆ Create consistent KPI calculation rules
- ◆ Maintain shared dimensions for employees, teams, sites, and processes

Automated Data Quality Controls

Implement automated validation rules including:

- ◆ Completeness checks
- ◆ Duplicate detection
- ◆ Referential integrity validation
- ◆ Data volume monitoring
- ◆ Business rule validation
- ◆ Timestamp consistency checks

Data Quality Monitoring

- ◆ Create data quality dashboards
- ◆ Generate alerts for abnormal data patterns
- ◆ Maintain data quality issue logs

Business Validation

- ◆ Conduct workshops with Finance Operations teams
- ◆ Validate KPI definitions and calculation logic
- ◆ Establish data ownership responsibilities

SOURCE SYSTEM AVAILABILITY RISKS

The solution depends on availability and reliability of multiple operational systems.

Potential risks include

- ◆ SAP ECC or SAP S/4HANA downtime
- ◆ Planned maintenance windows
- ◆ API/service interruptions
- ◆ Changes in authentication mechanisms
- ◆ Delayed availability of operational data

MITIGATION STRATEGY

Resilient Data Integration

- ◆ Monitoring of data pipelines
- ◆ Use retry mechanisms for failed loads
- ◆ Maintain detailed execution logs
- ◆ Enable failure notifications

Decoupled Architecture

- ◆ Separate data ingestion from reporting layers
- ◆ Store historical data independently from source systems
- ◆ Allow Power BI reporting to continue using the latest available dataset during temporary source outages

Operational Monitoring

Monitor:

- ◆ Pipeline execution status
- ◆ Data refresh completion
- ◆ Source system connectivity
- ◆ Data availability SLA

Coordination with Source System Owners

- ◆ Define integration ownership model
- ◆ Align refresh schedules with SAP maintenance windows
- ◆ Establish incident management procedures

SECURITY RISKS

The solution processes operational and employee-related information, requiring appropriate security controls.

Potential security risks include

- ◆ Unauthorized access to employee productivity information
- ◆ Excessive user permissions
- ◆ Exposure of sensitive operational data
- ◆ Incorrect role configuration
- ◆ Security issues related to APIs and service accounts

MITIGATION STRATEGY

Identity and Access Management

- ◆ Use Microsoft Entra ID for authentication and identity management
- ◆ Enable Single Sign-On (SSO)
- ◆ Apply Multi-Factor Authentication (MFA) according to corporate policies
- ◆ Use managed identities/service principals for automated processes

Authorization Controls

- ◆ Implement Role-Based Access Control (RBAC)
- ◆ Apply Power BI Row-Level Security (RLS)
Restrict access based on:
 - ◆ Country
 - ◆ Site
 - ◆ Company Code
 - ◆ Team
 - ◆ Management hierarchy

Data Protection

- ◆ Encrypt data at rest and in transit
- ◆ Secure API credentials using Azure Key Vault

Governance and Monitoring

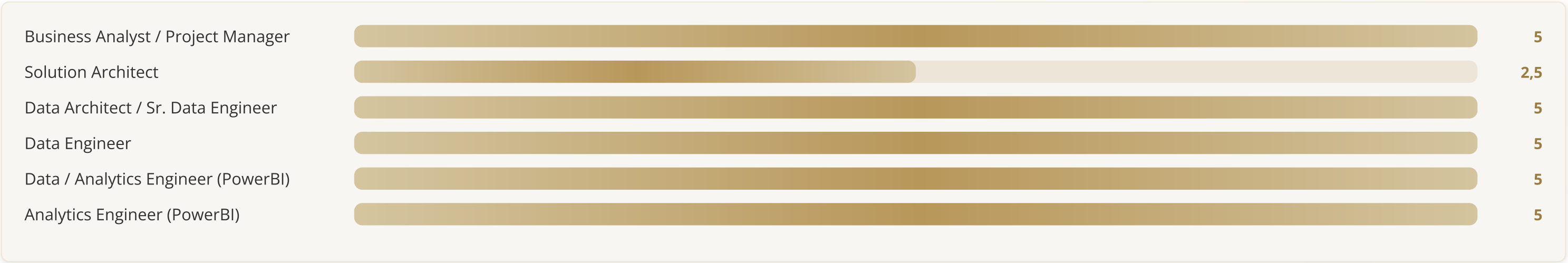
- ◆ Enable audit logging
- ◆ Monitor user access and report usage

CALCULATION

TEAM ALLOCATION – ROLES, MONTHS & EFFORT

Role	Month 1	Month 2	Month 3	Month 4	Month 5	Total MM	Total MD
Business Analyst / Project Manager	1	1	1	1	1	5	100
Solution Architect	0,5	0,5	0,5	0,5	0,5	2,5	50
Data Architect / Senior Data Engineer	1	1	1	1	1	5	100
Data Engineer	1	1	1	1	1	5	100
Data Engineer / Analytics Engineer (PowerBI)	1	1	1	1	1	5	100
Analytics Engineer (PowerBI)	1	1	1	1	1	5	100
Total	5,5	5,5	5	5	5,5	26,5	530

TOTAL MAN-MONTHS BY ROLE



26,5

TOTAL MAN-MONTHS

530

TOTAL MAN-DAYS

5

MONTHS DURATION

6

TEAM ROLES